From Detection to Attribution: Linking Two Decades of Antarctic Stream-Channel Change to Its Climate Drivers, from Terrain Data Alone

Summary. This project turns an existing Antarctic stream-change monitor into a tool that explains and predicts change. Barlow (2026) mapped two decades of stream-channel change across the McMurdo Dry Valleys; the proposed work asks the next questions. (O1) Which climate drivers control that change? (O2) Can the terrain-only detector work as well on free satellite data as it does on airborne lidar, across all four valleys? (O3) Can every change map carry an honest, per-pixel error bar? The measurement foundation is already running and reproduces Barlow's published uncertainty ranges, and a first driver signal is shown in §4.

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1. Background and the Gap

The McMurdo Dry Valleys (MDVs) are Earth's largest ice-free Antarctic desert and, for a century, were considered the planet's most geomorphologically stable landscape. That assumption is breaking down. The system is energy-limited — there is plenty of frozen water but barely enough heat to melt it — so even small increases in absorbed sunlight, together with thawing permafrost, translate directly into ephemeral summer streams: channels that flow for only a few weeks each year, move sediment while they do, and reshape themselves in the process. These streams are the most sensitive measurable indicator of climate change in the most stable place on Earth, a continental-scale "climate canary."

Barlow (2026) [1] established the measurement foundation in three transferable pillars:

1. Terrain-only semantic segmentation: a U-Net delineates stream boundaries from DEM derivatives alone (elevation, slope, aspect most informative; no water/color signal), so it works in channels that are dry 11 months a year. 2. Satellite-DEM validation: free REMA satellite DEMs are proven accurate enough for centimeter-to-meter stream-corridor change detection, using ICP (iterative closest point — sliding one 3-D surface onto another until they match) for alignment and NMAD (normalized median absolute deviation — a standard deviation that ignores outliers) for honest error bars; the errors follow a Laplacian (sharp-peaked, heavy-tailed) distribution rather than a Gaussian, which is why the robust statistic is required. 3. Multi-epoch change detection: subtract one epoch's DEM from another (DEM-of-Difference, DoD) inside the segmented stream masks, and count only elevation changes exceeding the level of detection (LOD95 = 1.96 × NMAD — the smallest change distinguishable from measurement noise at 95% confidence). This yields per-stream erosion/deposition and acceleration rates across four valleys and three epochs (2001 & 2014 lidar, 2021–23 REMA).

The gap. Barlow's dissertation ends at *description*: it maps where change is happening (Denton Hills is the erosion hotspot; parts of Taylor Valley are accelerating) but explicitly defers two threads to future work: (a) coupling geomorphic change to physical/climate drivers, and (b) generalizing the model beyond the MDVs. A Future Investigator project lives precisely in that gap: turning a *monitor* into an *explanatory and predictive* tool (Fig. 1). This is a genuine scientific increment, not a replication.

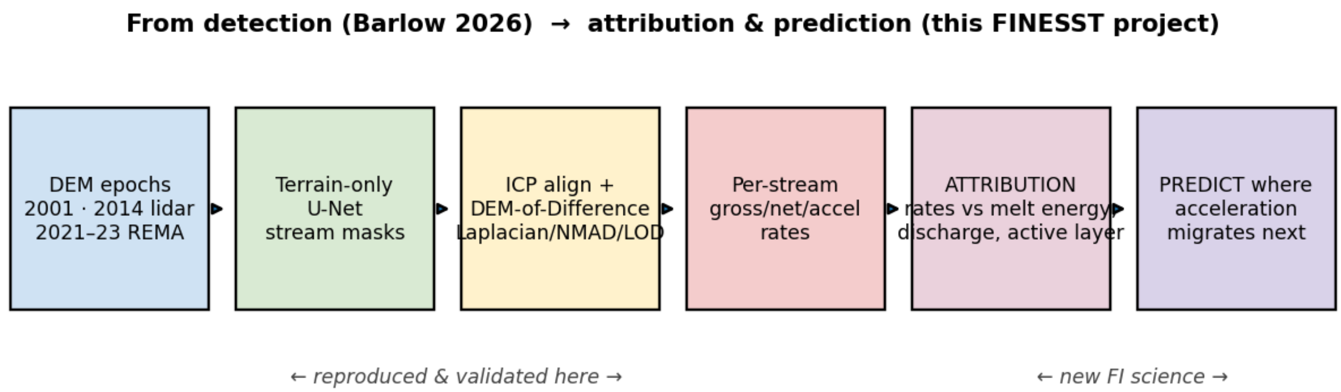

Figure 1. From detection (Barlow 2026; the first four stages) to the proposed attribution and prediction science (last two stages, the new contribution of this project).

Relevance to NASA. The 2001 baseline epoch is NASA data — an ATM airborne lidar campaign — that has never been fully exploited for surface change. The proposed work converts that NASA archive, the free REMA satellite record, and reanalysis into a continuing, uncertainty-calibrated measurement of climate-driven surface change in Earth's largest ice-free Antarctic region, serving the Earth Science Division's cryosphere and Earth-surface-change focus areas. The methodological deliverables generalize: terrain-only feature detection (O2) and calibrated per-pixel change thresholds (O3) apply to any elevation-differencing record, including those produced by ICESat-2 and NISAR. The MDVs are also the canonical terrestrial analog for cold-desert planetary surfaces — a long-standing NASA investment this project extends with open, reusable tools. The FI development itself is Division-aligned: open geospatial data, machine learning on remote sensing, and honest uncertainty quantification.

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2. Objectives and Hypotheses

#	Objective	Hypothesis	Increment over Barlow
O1	Attribution (headline science). Match each stream's measured change rate against candidate drivers — temperature (as positive degree-days, PDD: a running total of melting weather), sunlight, discharge, glacier mass balance, thaw depth — from the MDV-LTER network plus ERA5/AMPS.	H1: *cumulative melt energy*, not water volume alone, controls sediment movement — a heat-based model predicts each stream's rate better than discharge and explains the scatter discharge leaves behind.	Listed verbatim as Barlow future work; never quantified.
O2	Cross-sensor & cross-valley generalization. Measure exactly how satellite DEMs disagree with lidar (the domain shift), correct it, and show near-lidar detector performance on satellite data across all four valleys.	H2: most of the sensor gap is predictable, terrain-dependent bias; correcting it per pixel (slope, aspect) recovers near-lidar accuracy on REMA.	Barlow flags sensor generalization as future work.
O3	Calibrated uncertainty. Replace the single valley-wide noise estimate with a per-pixel threshold that accounts for slope, aspect, and sensor.	H3: a 95% threshold built this way behaves like one — on unchanged ground, only ~5% of pixels exceed it.	Barlow uses a single global NMAD per epoch-pair.

Predictive payoff (ties O1–O3 together): a model that, given a driver field, predicts *where acceleration migrates next*, the actionable form of the "climate canary."

3. Methodology

Data (all free/public; see §6). Three elevation snapshots form the backbone: 2001 airborne lidar (NASA ATM, 2 m), 2014 airborne lidar (NCALM, 1 m), and the REMA satellite DEM (2 m, 2021–23). From each snapshot the pipeline derives the terrain layers the detector reads: elevation, slope, aspect, curvature, flow accumulation (a "where water would go" map), and lidar intensity. Driver data come from the MDV-LTER station network — meteorology, 21 stream-discharge gauges, mass balance for 7 glaciers, and soil temperature/moisture (a proxy for summer thaw depth) — with the ERA5 and AMPS weather models filling the gaps between stations. Training labels: the public MCM-LTER stream centerlines, standing in for Barlow's access-gated 217 hand-digitized tiles (§7 risk).

O1, Attribution. For each gauged stream and epoch: (i) measure the change rate — the DoD inside its channel mask (already operational, Fig. 2); (ii) build the stream's energy history (PDD, sunlight, discharge, thaw depth) from LTER stations plus ERA5/AMPS; (iii) fit hierarchical regression and random-forest models relating rate to drivers, always validating on streams held out of fitting; (iv) test H1 head-to-head — does melt energy predict rates better than water volume alone? For the Taylor streams with all three epochs, the *change* in rate (acceleration) is compared against the *trend* in each driver.

O2, Generalization. The work will first measure how the satellite and lidar elevations disagree over ground that has not changed, and how that disagreement depends on slope and aspect. It will then learn a correction for the disagreement, retrain the segmenter with both sensors represented in its training data, and score accuracy (F1) across all four valleys and both sensor types. The FI's prior work applying this same architecture to a completely different landscape (§5) shows it transfers across sensors and biomes.

O3, Calibrated uncertainty. The work will replace the single global noise estimate with one that varies by slope, aspect, and sensor, and publish it as a per-pixel detection-threshold map. Calibration will be verified the honest way: on held-out ground that did not change, a 95% threshold should be exceeded by only ~5% of pixels. (Fig. 4 shows today's valley-wide version.)

Reproducibility/QC. The pipeline is fully scripted; input data and all derivatives regenerate from source, with robust statistics and CRS checks enforced at every step.

4. Preliminary Results

The change-detection foundation the proposed work builds on is already operational. A reproduction of Barlow's change detection on Taylor Valley falls inside her published uncertainty ranges (below), and an initial driver analysis surfaces the first attribution signal that O1 will develop.

(a) Change detection across all three epochs (Fig. 2).

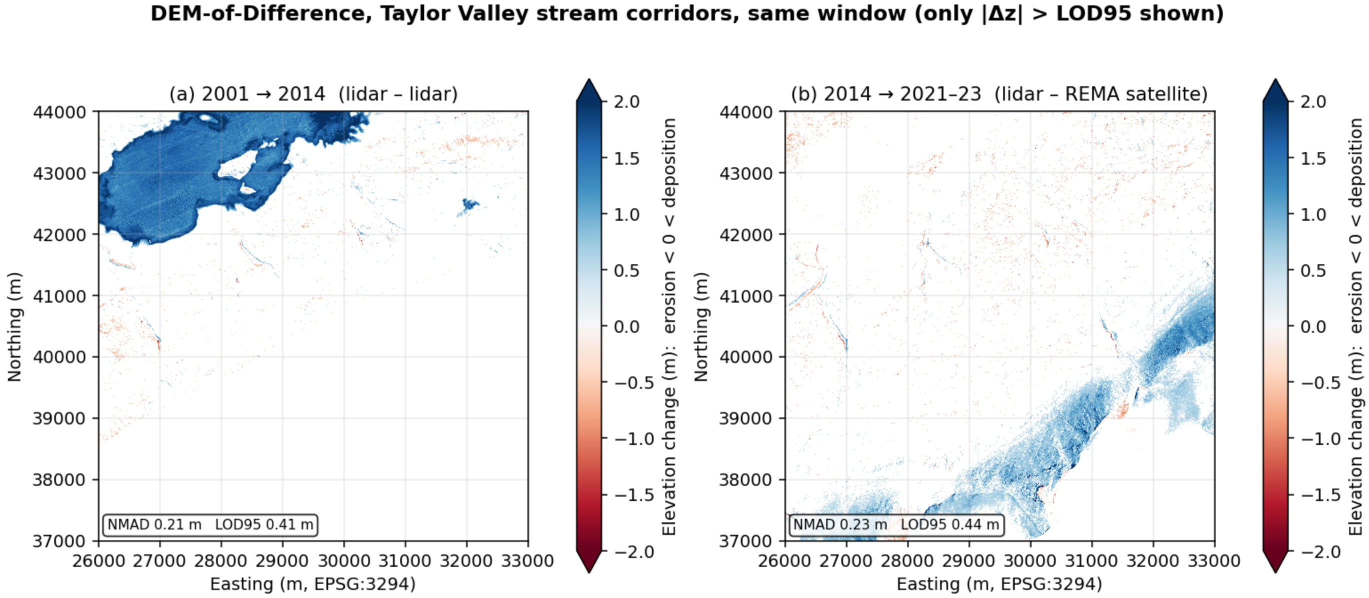

Figure 2. Elevation change over the same Taylor Valley window: (a) 2001→2014 (lidar–lidar); (b) 2014→2021-23 (lidar–REMA). Red = erosion, blue = deposition; only changes exceeding LOD95 are drawn. The flat patch in (a) is Lake Fryxell's ~1.5 m rise — an automated standing-water screen detects and excludes such surfaces from every stream-rate statistic.

Epoch pair	NMAD	LOD95	Barlow's published range	In range?
2001→2014 (lidar–lidar)	0.21 m	0.41 m	NMAD 0.07–0.46 / LOD95 0.15–0.92	✓
2014→2021-23 (lidar–REMA)	0.23 m	0.44 m	NMAD 0.19–0.53 / LOD95 0.37–1.04	✓

ICP alignment behaved exactly as Barlow's published version does: it tightened the error budget on steep windows and correctly added no benefit on the flat valley floor.

(b) Per-stream rates (Fig. 3), the masked products O1 consumes.

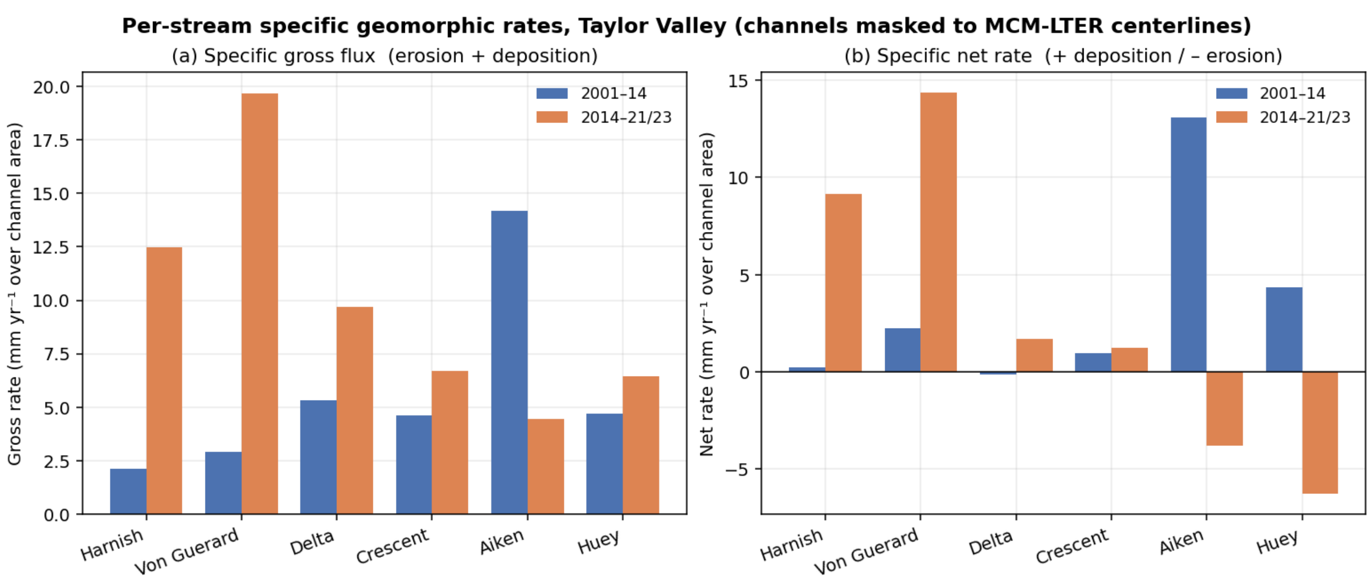

Figure 3. Per-stream sediment rates for six gauged Taylor Valley streams, reported per square meter of channel (mm/yr): the survey footprints differ across epochs, and a per-area rate cannot be inflated simply because one survey saw more ground.

(c) Robust error model (Fig. 4), the basis for O3.

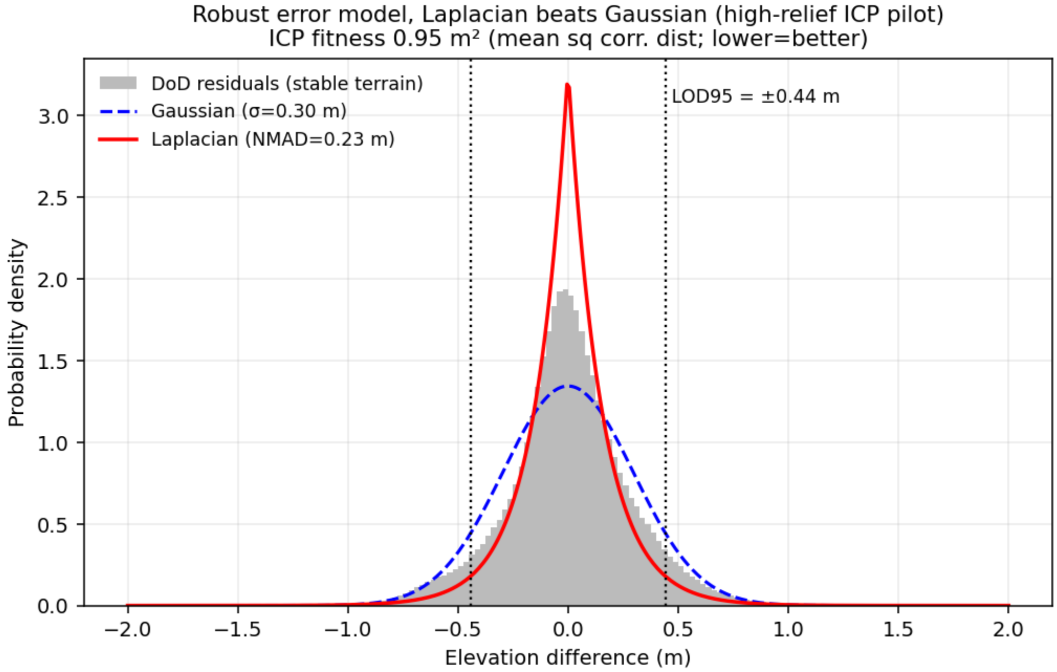

Figure 4. Measurement noise profiled on unchanged ground: sharply peaked with heavy tails (Laplacian), not Gaussian — outliers would inflate a naive standard-deviation threshold, hence NMAD. Reproduces Barlow's error-model finding; launch point for O3.

(d) A first attribution signal (Fig. 5).

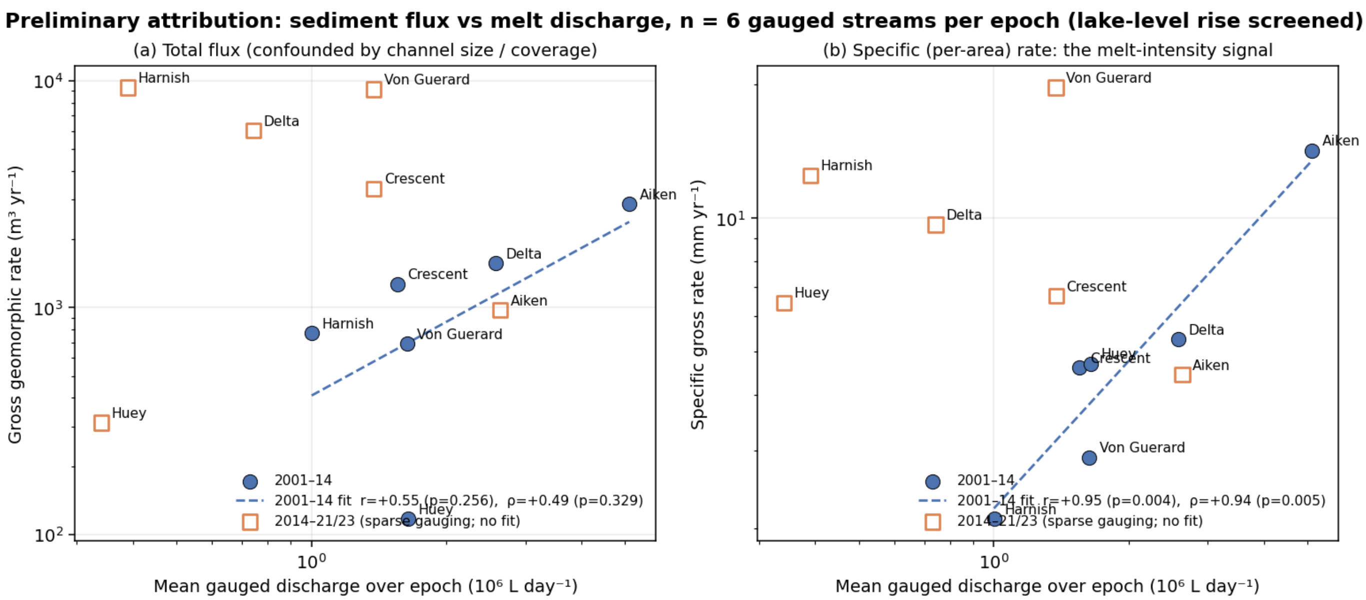

Figure 5. The first attribution signal: per-stream sediment rate vs mean gauged discharge (log-log; lake signal screened). In 2001→2014 the relationship is strong — r = +0.95 (p = 0.004), ρ = +0.94 (p = 0.005) — and survives dropping any single stream. The satellite period shows no coherent relation for an instructive reason: post-2015 gauging is sparse (48–362 recorded days per stream), so no fit is drawn. That is the O1 motivation — replace patchy gauging with modeled melt energy to extend attribution across all epochs and valleys.

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5. FI Qualifications

The FI has independently built and operated this exact toolchain, U-Net semantic segmentation on lidar-derived terrain rasters, ICP co-registration, and DEM-of-Difference change analysis, on an unrelated landscape (channels, roads, and disturbance scars detected from elevation alone in the Appalachian Plateau). That prior work shows the architecture transfers across sensors and biomes, directly de-risking O2, and that the FI already commands the full pipeline end to end.

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6. Data Requirements and Availability

Every dataset required for the proposed work is free/public and already in hand (regenerable from source via scripted pipelines), so data availability poses no schedule risk. The single access-gated dependency is Barlow's training labels (mitigated below and in §7).

Dataset (source)	What it shows	Role in the proposed work	Status
2001 airborne lidar, 2 m (NASA ATM, via USGS)	Valley-floor surface in 2001	Baseline epoch for all change detection	✓ in hand
2014 airborne lidar, 1 m (NCALM, via OpenTopography)	Same terrain 13 yr later at benchmark accuracy	Reference epoch; anchors the error model	✓ in hand
REMA satellite DEM, 2 m, 2021-23 (PGC/Maxar stereo)	Most recent surface, continent-wide	Extends the record past the last lidar flight; the sensor O2 must generalize to	✓ in hand
Per-epoch terrain derivatives (slope/aspect/curvature/MFD-flow/intensity)	Channel-diagnostic terrain form	Input features the U-Net segmenter reads	✓ scripted; demonstrated (Taylor pilot)
LTER stream gauges, 21 streams, daily, 1990s-present (EDI)	Meltwater each stream carried	Direct discharge driver for O1 (source of the Fig. 5 signal)	✓ in hand
LTER meteorology (EDI)	Air temperature, radiation, wind	PDD + insolation melt-energy drivers (O1/H1)	✓ in hand
LTER glacier mass balance, 7 glaciers (EDI)	Seasonal ice gain/loss per glacier	Links stream water supply to its source	✓ in hand
LTER soil temperature/moisture (EDI)	Summer thaw depth	Active-layer driver (O1/H1)	✓ in hand
ERA5 monthly 1993-2024 (Copernicus CDS); AMPS 2.67 km (NCAR GDEX)	Continuous modeled atmosphere	Gap-fills drivers between stations and to ungauged streams; AMPS cross-checks ERA5	✓ in hand
MCM-LTER stream channel polygons (EDI `knb-lter-mcm.6007`)	Mapped channel of every named stream	Per-stream rate masks (used in Figs. 3/5) + label stand-in	✓ in hand
Barlow's 217 hand-digitized tiles (author-gated)	Expert-traced channel boundaries	Gold-standard U-Net training labels	▲ lone access-gated dependency (see §7)
Cross-valley DEMs, all four valleys (OpenTopography/PGC)	Terrain beyond Taylor Valley	O2 generalization test bed	✓ DEMs in hand; per-valley stacks to build
WorldView/Maxar imagery; published rates (PGC restricted; literature)	Visual ground truth; independent rates	Optional label QC / cross-check vs Barlow	optional

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7. Management, Timeline, Risks, and Data Management Plan

FI role & development. The FI is intellectual lead and primary author; the advisor (PI of record) provides domain mentorship. Development plan: present at AGU (cryosphere) and an ISAES/SCAR venue; co-author the attribution paper (O1) as a stand-alone publication; complete graduate coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Period	Milestones
Yr 1	Full driver harmonization (PDD/insolation/active-layer series per gauged stream); O1 attribution model on Taylor Valley (extending the Fig. 5 signal); calibrated-uncertainty prototype (O3). Deliverable: attribution manuscript drafted.
Yr 2	Cross-sensor domain-shift correction + segmenter generalization across all four valleys (O2); apply attribution model basin-wide; AGU presentation. Deliverable: O2 paper.
Yr 3	Predictive model (where acceleration migrates next); 3-epoch acceleration attribution; calibrated-uncertainty product release (O3). Deliverable: synthesis/prediction paper + public data products.

Risks & mitigations. (1) *Barlow's label tiles are access-gated* → mitigation: LTER centerlines already work as a QC stand-in (Fig. 2–3 built on them); plan to request the tiles and, failing that, re-digitize a comparable training set (the FI's prior work shows label generation is in hand). (2) *REMA's uneven post-2014 coverage* → prioritize Taylor Valley (full 3-epoch coverage) for acceleration; treat other valleys as 2-epoch. (3) *Discharge gauge gaps* → ERA5/AMPS energy reanalysis fills spatial/temporal holes (a core reason O1 uses melt energy, not just gauges).

Data Management Plan. All input data are free/public (NASA ATM, NCALM/OpenTopography, PGC REMA, MCM-LTER/EDI, Copernicus ERA5, AMPS/GDEX), cited per source. Derived products, change-detection rasters, per-stream rate tables, calibrated-uncertainty layers, and stream-boundary polygons, will be released open-access (DoIs via Zenodo / EDI) with the processing code (PDAL/WhiteboxTools/GDAL + Python). Heavy regenerable rasters are excluded from version control; lightweight scripts, figures, and tables are tracked. Outputs follow ASPRS/OGC standards and carry explicit CRS, method, and uncertainty metadata (per NASA open-science/trustworthy-EO guidance).

References

[1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica.* PhD Dissertation, Dept. of Civil & Environmental Engineering, University of Houston (chair: C. L. Glennie). 240 pp.

[2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234. doi:10.3390/rs14010234. *(Dissertation Chapter 4, proof of concept.)*

[3] Höhle, J., & Höhle, M. (2009). Accuracy assessment of digital elevation models by means of robust statistical methods. *ISPRS J. Photogramm. Remote Sens.* 64(4):398–406. *(NMAD / robust DEM error.)*

[4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674. *(REMA.)*

[5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.

[6] McMurdo Dry Valleys LTER. Stream discharge, meteorology, glacier mass-balance, and soil datasets, Environmental Data Initiative (EDI), `knb-lter-mcm.*`.

[7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Draft S/T/M section. Figures generated by `barlow/build/_finesst_figures.py`*. Not a submitted proposal.*